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RV COLLEGE OF ENGINEERING®

Autonomous Institution affiliated to VTU

IV Semester B. E. Examinations
ELECTRONICS AND INSTRUMENTATION ENGINEERING
Model Question Paper
EI243AI – Microcontroller & Programming

Time: 03 Hours **Maximum Marks: 100**

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 to 10.

PART-A

			M	BT	CO
1	1.1	After performing data conversion operation, the contents in register R1 was found to be R1=0x00000000. The contents in R0 register was initially R0=0x55AA8700.Identify the instruction was executed.	2	3	2
	1.2	_____________ Instructions provide support for subroutine entry and exit.	1	1	1
	1.3	Each instruction in Cortex-M machines is encoded into ____ bytes word.	1	1	1
	1.4	How many bits data bus is used in Cortex?	1	1	1
	1.5	Which of the following instructions are accessing the memory locations? i) Store ii) MOVE iii) Load iv) arithmetic v) logical	2	2	2
	1.6	How many instructions pipelining is used in Cortex-M?	1	1	1
	1.7	Identify the instructions which are called Program Status Register transfer instructions.	2	2	1
	1.8	Write an assembly instruction would you use to load 4 words starting from the memory location 0x80000000 into the registers r0-r3? (Assume r9 contains the base address 0x80000000).	2	2	2
	1.9	When building code for different cortex states, ___________tool decides for each function call whether to use a BL or BLX instruction.	1	1	1
	1.10	ADD r3, r2, r1, LSL #3; instruction performs _______________.	1	2	2
	1.11	While CPU is executing a program, an interrupt exists then it ______ the normal sequence of execution of instructions	1	1	1
	1.12	In the branch instructions of Cortex, what does the mnemonic BVC	1		

		imply?		1	1
	1.13	In LCD, which function is executed by ‘0x05’ hex command?	1	1	1
	1.14	In DC motor interfacing, which field/s is/are generated by forcing current through the coil for spinning of the motor?	1	2	2
	1.15	In DC motor interfacing, which modulation controls the duty cycle of square wave provided at the output by generating variation in the average DC voltage?	1	2	2
	1.16	In USART-CR1 register if over8 bit is ‘0’, it indicates-------------	1	1	1

PART-B

2	a	Differentiate between RISC and CISC architectures and give suitable example for each.	6	1	1
	b	What is the significance of floating-point arithmetic? And represent **$(1023.125)_{10}$** in single and double precision formats	6	3	2
	c	Highlight the features of STM32F407 Processor.	4	1	1

3	a	What are data processing instructions in STM32F407 processor? Define any ten data processing instructions and illustrate with an example for each.	6	2	2
	b	Illustrate the multiple memory loading and store instructions with an example for each.	6	2	2
	c	Write an ARM Cortex-M Assembly Language Program to transfer the 10, 8-bits block of data from block of memory locations **X to Y.**	4	3	2
OR					
4	a	Write an ARM Cortex-M Assembly language program to find the number of bits set in a 32-bit data, given by the user at 0x40000000 memory location. Store the result in 0x40000050, memory location.	8	3	3
	b	With a neat block diagram, discuss ARM Cortex-M core data flow model.	8	2	2

5	a	Describe the ARM Cortex Memory organization with neat diagram	6	2	2
	b	Show how the processing time is reduced in Cortex-M compared to basic ARM instructions.	4	2	2
	c	Write a complete ARM Cortex-M **‘C’ program** for while loop using CubeMX tool for **Potentiometer interfacing** to perform **ADC**. Draw the interfacing diagram. Also write an algorithm for the same	6	3	3
OR					
6	a	Write a complete ARM Cortex-M **‘C’ program** for while loop using CubeMX tool for Push button interfacing. Use blue LED**.** Also write an algorithm for the same.	8	3	3
	b	Write a complete ARM Cortex-M **‘C’ program** for while loop using CubeMX tool for **Potentiometer interfacing** to perform **DAC**. Draw the interfacing diagram. Also write an algorithm for the same	8	3	3

7	a	Write a **‘C’ program** using CubeMX tool to illustrate the USART transmission.	8	3	3

	b	Write a **‘C’ program** using CubeMX tool to demonstrate the SPI data transfer.	8	3	3
		OR			
8	a	Write an **‘C’ program** using CubeMX tool to configure the following: i) If PD.8 is set, then to rotate the stepper motor in clockwise rotation. else anticlockwise rotation. ii) PD10 to control the speed of the motor.	8	4	3
	b	Configure the Seven segment display of ARM STM32F4XX board to display decimal digits and write a **‘C’ program** using CubeMX tool	8	4	3
9	a	Illustrate the operation of timers available in Cortex Processor.	8	2	2
	b	Write a **‘C’ program** using CubeMX tool to demonstrate the operation of pulse width modulation.	8	3	2
		OR			
10	a	List the various interrupt handling schemes. With an example clearly highlight the mechanism of nested vector interrupt handler.	8	2	2
	b	Describe the following interrupt registers i. PRIMASK ii. FAULTMASK and BASEPRI along with their register configuration	8	2	2